drafting-viz

```
# Load necessary libraries
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.5
✓ forcats   1.0.0     ✓ stringr   1.5.1
✓ ggplot2   4.0.1     ✓ tibble    3.2.1
✓ lubridate 1.9.4     ✓ tidyr     1.3.1
✓ purrr     1.0.4
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
library(here)
```

```
here() starts at C:/Users/oc3an/OneDrive/Documents/MEDS/EDS-240/eds240-infographic
```

```
library(janitor)
```

```


Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(ggspatial)
library(prettymapr)
library(showtext)
```

```
Loading required package: sysfonts
Loading required package: showtextdb
```

```
font_add_google(name = "Noto Sans", family = "noto_sans")
font_add_google("Libre Baskerville", family = "libre")
```

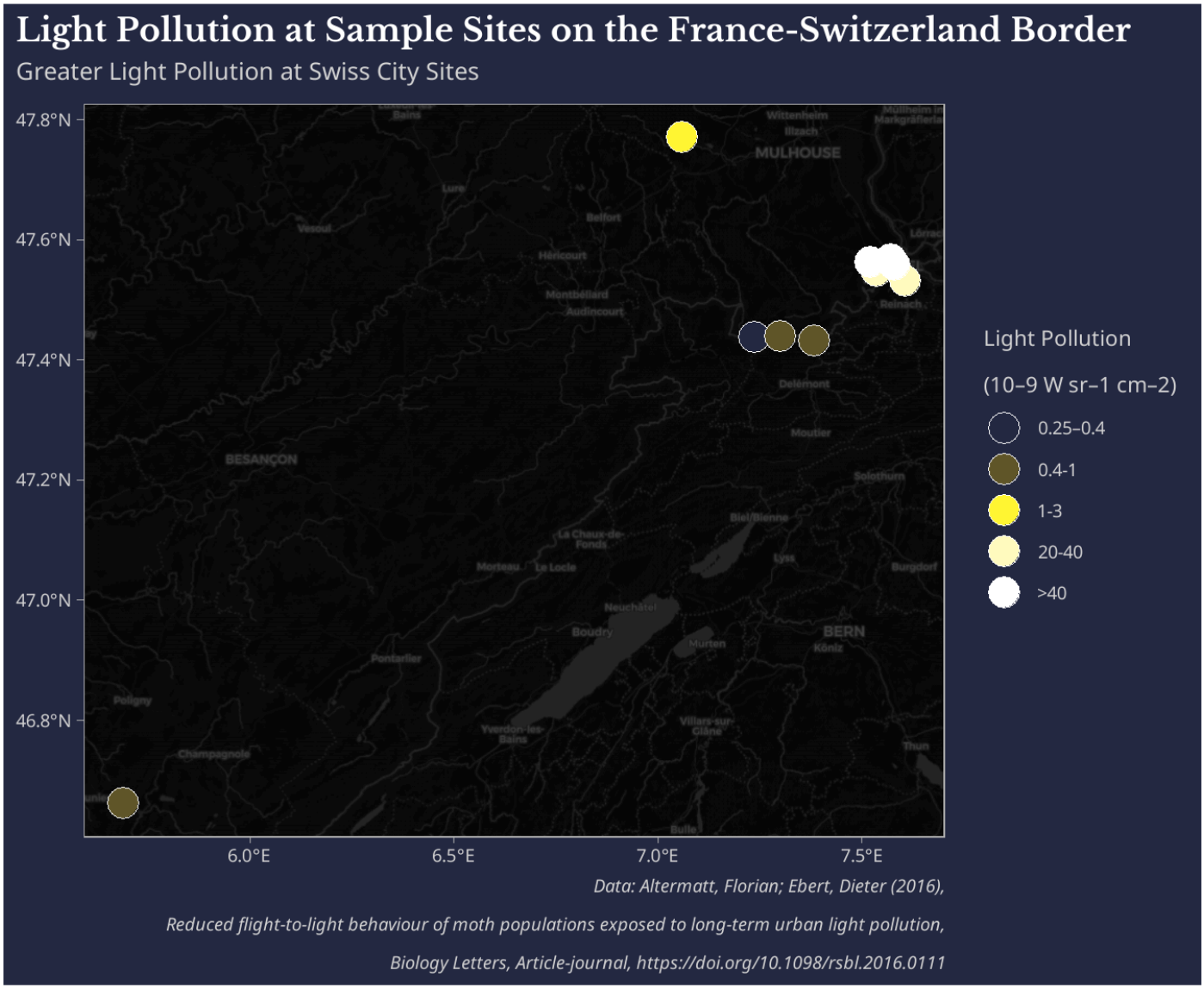
```
# Load in data and standardize column names
light_pol <- read.delim(here("data", "light_pol.txt"), skip = 2) %>%
```

```r
  clean_names() %>%
  # Separate location column
  separate_wider_delim(
    cols = location_country,
    delim = "/",
    names = c("loc_name", "country")
  ) %>%
  # Remove any whitespace from separated columns
  mutate(country = str_squish(country), loc_name = str_squish(loc_name)) %>%
  # Rename population column to be more relevant
  rename(pollution_status = population) %>%
  # Recode values to be more intuitive
  mutate(
    pollution_status = recode(
      pollution_status,
      "dark-sky" = "non polluted",
      "light-polluted" = "polluted"
    ),
    # Recode country codes to full names
    country = recode(country, "F" = "France", "CH" = "Switzerland"),
    # Fix location name with conversion error
    loc_name = recode(loc_name, "Kleinl⍰tzel" = "Kleinlützel")
  ) %>%
  # Make a new column with pollution bin values
  mutate(
    pollution_bin = case_when(
      loc_name == "Blochmont"       ~ "0.25–0.4",
      loc_name == "Kleinlützel"     ~ "0.4-1",
      loc_name == "Kiffis"          ~ "0.4-1",
      loc_name == "Doucier"         ~ "0.4-1",
      loc_name == "Lutterbach"      ~ "1-3",
      loc_name == "Allschwil"       ~ "20-40",
      loc_name == "Reinach"         ~ "20-40",
      loc_name == "Hegenheim"       ~ ">40",
      loc_name == "Basel Kannenfeld" ~ ">40",
      loc_name == "Basel Spalentor" ~ ">40"
    ),
    # New Latitude and Longitude column with values from paper
    lat = case_when(
      loc_name == "Blochmont"        ~ 47 + 26 / 60 + 16 / 3600,
      loc_name == "Kleinlützel"      ~ 47 + 25 / 60 + 55 / 3600,
      loc_name == "Kiffis"           ~ 47 + 26 / 60 + 18 / 3600,
      loc_name == "Doucier"          ~ 46 + 39 / 60 + 40 / 3600,
      loc_name == "Lutterbach"       ~ 47 + 46 / 60 + 12 / 3600,
      loc_name == "Allschwil"        ~ 47 + 32 / 60 + 52 / 3600,
      loc_name == "Reinach"          ~ 47 + 31 / 60 + 50 / 3600,
      loc_name == "Hegenheim"        ~ 47 + 33 / 60 + 43 / 3600,
      loc_name == "Basel Kannenfeld" ~ 47 + 33 / 60 + 59 / 3600,
      loc_name == "Basel Spalentor"  ~ 47 + 33 / 60 + 28 / 3600
    ),
    lon = case_when(
```

```
      loc_name == "Blochmont"        ~  7 + 14 / 60 + 10 / 3600,
      loc_name == "Kleinlützel"      ~  7 + 22 / 60 + 57 / 3600,
      loc_name == "Kiffis"           ~  7 + 17 / 60 + 59 / 3600,
      loc_name == "Doucier"          ~  5 + 41 / 60 + 21 / 3600,
      loc_name == "Lutterbach"       ~  7 +  3 / 60 + 36 / 3600,
      loc_name == "Allschwil"        ~  7 + 32 / 60 +  8 / 3600,
      loc_name == "Reinach"          ~  7 + 36 / 60 + 25 / 3600,
      loc_name == "Hegenheim"        ~  7 + 31 / 60 + 14 / 3600,
      loc_name == "Basel Kannenfeld" ~  7 + 34 / 60 + 19 / 3600,
      loc_name == "Basel Spalentor"  ~  7 + 34 / 60 + 53 / 3600)) %>%
  mutate(pollution_bin = factor(pollution_bin, levels = c("0.25–0.4",
                                                          "0.4-1",
                                                          "1-3",
                                                          "20-40",
                                                          ">40")
                                )
         )

# Create sf object for mapping
light_pol_sf <- st_as_sf(light_pol, coords = c("lon", "lat"), crs = 4326)
```

```
# caption <- glue::glue(
#   "Data: Altermatt, Florian; Ebert, Dieter (2016), Reduced flight-to-light behaviour of moth po
#   Biology Letters, Article-journal, https://doi.org/10.1098/rsbl.2016.0111"
# )

showtext_auto(enable = TRUE)

ggplot(light_pol_sf) +
  annotation_map_tile(type = "cartodark", zoomin = 0) +
  geom_sf(
    aes(fill = pollution_bin),
    color = "white",
    size = 6,
    shape = 21
  ) +
  scale_fill_manual(values = c("#242842", "#625929", "#FFF533", "#FFFCC2", "white")) +
  labs(title = "Light Pollution at Sample Sites on the France-Switzerland Border",
       subtitle = "Greater Light Pollution at Swiss City Sites",
       caption = "Data: Altermatt, Florian; Ebert, Dieter (2016),\n Reduced flight-to-light behav
  theme_light() +
  guides(fill = guide_legend(
    # Legend title
    title = "Light Pollution\n(10–9 W sr–1 cm–2)"
  )) +
  theme(
    plot.background = element_rect(fill = "#242842"),
    panel.background = element_rect(fill = "#242842"),
    legend.background = element_rect(fill = "#242842"),
```

```
  # Align with y axis text instead of just axis
  plot.title.position = "plot",

  plot.title = element_text(
    face = "bold",
    family = "libre",
    # Size relative to base size in theme
    size = 27,
    lineheight = 1.5,
    color = "white"),

  plot.subtitle = element_text(
    family = "noto_sans",
    size = 20,
    margin = margin(b = 8),
    color = "lightgrey"),

  # Add margin space below
  axis.text = element_text(size = 15,
    family = "noto_sans",
    color = "lightgrey"),

  plot.caption = element_text(
    family = "noto_sans",
    face = "italic",
    size = 15, color = "lightgrey"),

  legend.text = element_text(size = 15,
    family = "noto_sans", color = "lightgrey"),
  legend.title = element_text(size = 18,
    family = "noto_sans", color = "lightgrey")
  )
```




```
# Figure out how to fix alpha somehow
```

```
ggsave(here("figures", "draft_map.png"),
                      device = "png",
                      width = 8,
                      height = 7,
                      units = "in",
                      dpi = 300)
```

```
Zoom: 9
```

ideas - donut plot somehow - proportion of moths attracted to light in polluted vs non polluted areas - comparison of trends in male vs female moths to show any weird evolutionary trends between sexes

Implement other data? - show increases in light pollution over the last 10 years in Europe - display some visual for the importance of nocturnal pollinators

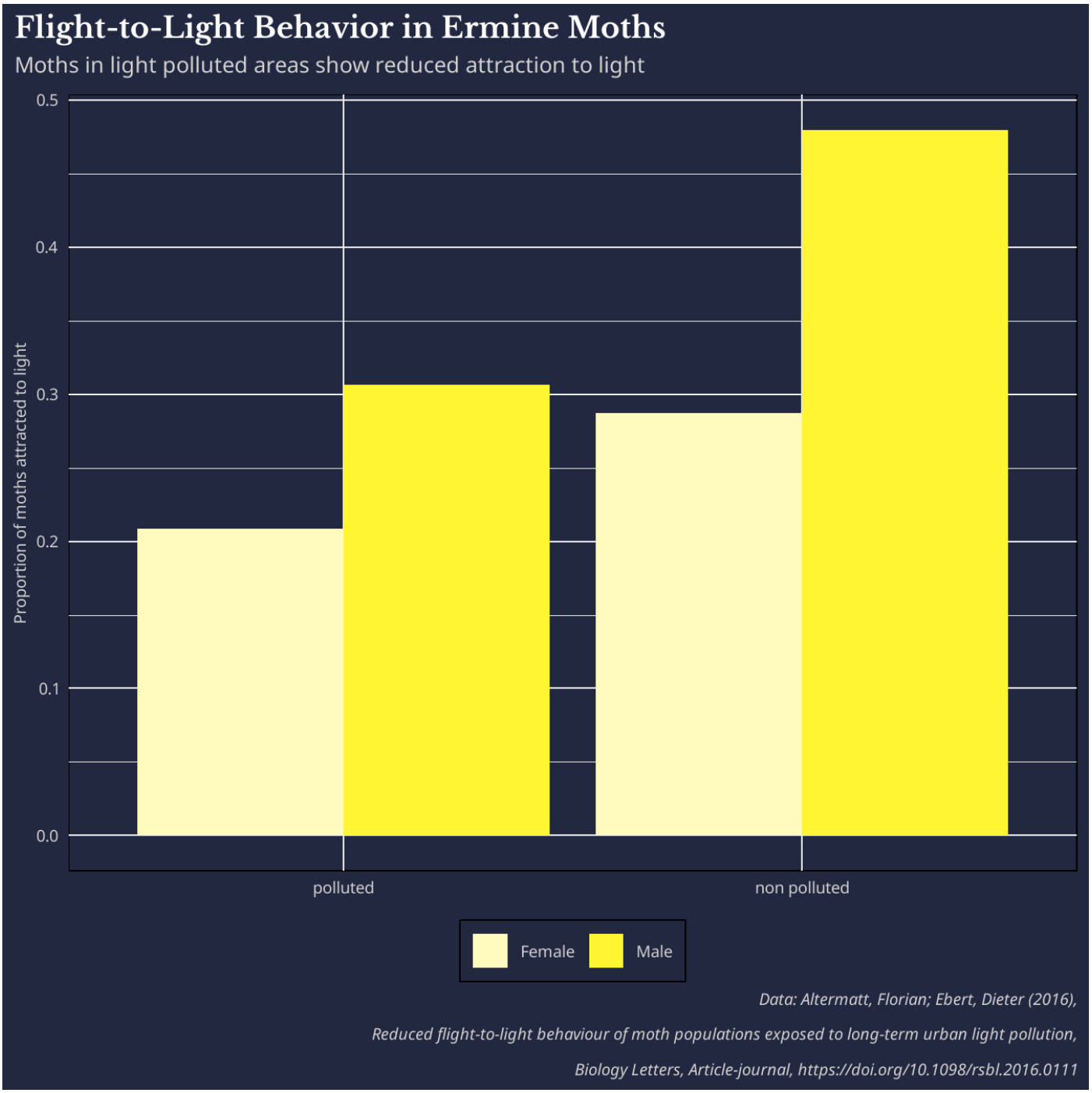
I want to find the proportion of moths that are attracted to light in “polluted” vs. “non-polluted” areas, how would I do this given the structure of the data?

```
showtext_auto(enable = TRUE)

pol_sum <- light_pol %>%
  group_by(pollution_status, sex) %>%
  summarise(
    total_attracted     = sum(moths_attracted),
    total_not_attracted = sum(moths_not_attracted),
    prop_attracted      = total_attracted / (total_attracted + total_not_attracted)
  ) %>%
  mutate(pollution_status = factor(pollution_status, levels = c("polluted", "non polluted")),
         sex = recode(sex,
                      "m" = "Male",
                      "f" = "Female"))
```

```
`summarise()` has grouped output by 'pollution_status'. You can override using
the `.groups` argument.
```

```
ggplot(pol_sum, aes(x = pollution_status, y = prop_attracted, fill = sex)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("#FFFCC2", "#FFF533")) +
  labs(title = "Flight-to-Light Behavior in Ermine Moths",
       subtitle = "Moths in light polluted areas show reduced attraction to light",
       caption = "Data: Altermatt, Florian; Ebert, Dieter (2016),\n Reduced flight-to-light behav
       y = "Proportion of moths attracted to light")+
  theme_minimal() +
    theme(
      legend.position = "bottom",
      axis.title.x = element_blank(),
      plot.background = element_rect(fill = "#242842"),
      panel.background = element_rect(fill = "#242842"),
      legend.background = element_rect(fill = "#242842"),

      # Align with y axis text instead of just axis
      plot.title.position = "plot",

      plot.title = element_text(
        face = "bold",
        family = "libre",
        # Size relative to base size in theme
        size = 27,
```

```
    lineheight = 1.5,
    color = "white"
  ),

  plot.subtitle = element_text(
    family = "noto_sans",
    size = 20,
    margin = margin(b = 8),
    color = "lightgrey"
  ),

  axis.title = element_text(
    size = 15,
    family = "noto_sans",
    color = "lightgrey"
  ),

  # Add margin space below
  axis.text = element_text(
    size = 15,
    family = "noto_sans",
    color = "lightgrey"
  ),

  plot.caption = element_text(
    family = "noto_sans",
    face = "italic",
    size = 15,
    color = "lightgrey"
  ),

  legend.text = element_text(
    size = 15,
    family = "noto_sans",
    color = "lightgrey"
  ),
  legend.title = element_blank()
  )
```


Flight-to-Light Behavior in Ermine Moths
Moths in light polluted areas show reduced attraction to light
Proportion of moths attracted to light
0.5
0.4
0.3
0.2
0.1
0.0
polluted
non polluted
Female
Male
Data: Altermatt, Florian; Ebert, Dieter (2016),
Reduced flight-to-light behaviour of moth populations exposed to long-term urban light pollution,
Biology Letters, Article-journal, https://doi.org/10.1098/rsbl.2016.0111